

Track 03

Advanced

The Architecture Track

G42 | Final: Friday 12 June 2026, Abu Dhabi.

A senior full-stack agentic AI track. You ship a working agent for two real dispute scenarios, with least-privilege tools, a fraud pause, and an audit trail the judges can actually read.

AT A GLANCE

Item	Detail
Who it is for	Senior engineers who are comfortable with Python and agent frameworks
Theme	AI agent architecture for the UAE
Partner	G42
Format	Working prototype, architecture documentation, and a 5-minute technical pitch
You deliver	A working agent for two dispute scenarios, with a readable audit trail and a human-approval gate
Pitch time	You build it on the day of the final
Tools	LangGraph, AutoGen, CrewAI, Python, and G42 APIs (alternatives and sandboxes are listed inside)

THE CHALLENGE: AGENTIC PAYMENTS, BILLING AND CONSUMER DISPUTE RESOLUTION

Payment, billing, and subscription disputes are slow, high-risk, and easy to get wrong. Refund a fraudster and you lose real money. Refuse a genuine customer and you lose their trust. Most support bots are just chatbots: they answer questions, but they cannot investigate a case, apply policy, or take a safe action. This track asks for something different. You build an agent that reasons, uses tools, holds state, and knows when to stop and ask a human.

Build an agentic AI that resolves payment, billing, subscription, refund, chargeback, and dispute issues end to end. It investigates the case, applies policy, proposes a remedy, and acts only with human approval when there is risk. It also catches fraudulent or abusive requests before any sensitive action runs. Pick from these domains: e-commerce, subscriptions, telecom, fintech, payments, and marketplaces.

This has to be agentic, not a chatbot. The agent is goal-directed, tool-using, stateful, grounded in policy, auditable, and risk-aware. It plans a sequence of steps, calls tools in order, carries state from one turn to the next, ties every remedy back to a versioned policy, and writes an audit log a human can read afterward.

Use mock data only. No live financial systems. What matters here is the architecture and the decision loop, not a payment integration. In your pitch, be straight about what is real and what is mocked.

WHAT YOU BUILD ON THE DAY

- A working agent that handles TWO scenarios. Scenario A is a suspicious refund request: a high-value 'item not received' claim, but delivery is confirmed and the account changed recently. The agent pauses, verifies, and routes the case to human review with an evidence summary. Scenario B is a duplicate or unrecognized charge: the agent matches the payment to the order, tells an auth hold apart from a settled duplicate, then either refunds or explains.
- The full loop for each scenario: verify identity (mocked), pull the evidence, screen for fraud risk, apply a versioned policy, propose a remedy, pause and route to a human-approval gate when there is risk, then write an audit log.
- 3 to 5 least-privilege tools, with read, prepare, and execute kept separate. For example: `get_order`, `get_payments`, `get_shipment`, `create_risk_case`, `prepare_refund`. Run the amount, policy, and idempotency checks before any prepare-refund.
- A versioned policy that the agent grounds its remedies in, plus eligibility checks against the facts of the case. Ground that policy in UAE consumer-protection principles (Federal Law No. 15 of 2020 on Consumer Protection); a full legal buildout is not required on the day.
- An audit trail anyone can follow: tool calls, evidence, risk score, decision rule, approval, and the customer notice.
- A 5-minute live technical demo of both journeys and the audit trail, plus a short pitch that is honest about what is real and what is mocked.
- Architecture documentation: the agent loop, the tools and their privileges, the policy model, and where the human-approval gate sits.

Out of scope for the day

- Live payment integration
- A full compliance buildout
- A production eval harness
- Voice
- Multi-channel intake

HOW YOU ARE JUDGED (100 POINTS)

A credible finalist scores 60 or above, with a working agent loop and a clear fraud pause. Two or three judges score independently, and your team score is the average of theirs.

Criterion (weight)	Excellent	Good	Needs work
Working agent loop (25)	An autonomous multi-step loop: the agent plans, calls tools in sequence, holds state across turns, and	Multi-step but brittle: some steps are hard-coded and state is limited.	Single-turn or scripted, with no real planning or state.

	recovers when a tool fails or returns nothing.		
Fraud detection + human-in-the-loop (20)	Catches risk signals, pauses the autonomous action, routes the case to a human-approval gate, and never accuses the customer without evidence.	Catches some risk and flags it, but the pause or the approval step is weak or entirely manual.	No risk screening, or the agent runs a sensitive action without checking.
Tool-call correctness + least-privilege (15)	The right tools with the right parameters; read, prepare, and execute are separated; and amount, policy, and idempotency checks run before any prepare-refund.	Mostly correct, with partial separation and some checks in place.	Wrong or over-privileged tools, and money actions run with no checks.
Auditability (15)	The full trajectory is shown in the demo: tool calls, evidence, risk score, decision rule, approval, and customer notice.	A partial log, with parts of the chain missing.	Little or no audit trail.
Policy grounding + remedy correctness (10)	The remedy follows a versioned policy, fits the facts of the case, and eligibility is checked.	A reasonable remedy, with the policy applied loosely.	An arbitrary or incorrect remedy.
Demo clarity + honesty (10)	A clear five-minute demo of both journeys, explicit about what is real and what is mocked, with a crisp narrative.	The demo works but is unclear or overclaims.	A confusing or non-working demo.
Innovation / stretch (5)	A real extra: a second domain, prompt-injection defense, a trace or eval metric, or a fresh take on the UX.	A minor extra.	None.

Tie-breaker: A tie is broken first by the Working agent loop score.

Tie-breaker: If you are still tied, the higher Fraud detection + human-in-the-loop score wins.

HOW TO WIN

- Get the loop running end to end on Scenario B first. A duplicate-charge case is easier to close than the suspicious refund, and a working loop is worth 25 points before anything else.
- Separate read, prepare, and execute from the start. Judges look for the amount, policy, and idempotency checks running before any prepare-refund. Build the gate in, do not bolt it on later.
- Make the fraud pause obvious in the demo. Show the risk signal firing, the autonomous action pausing, and the case routing to a human with an evidence summary. Never let the agent accuse the customer.
- Log everything as you go and put it on screen: tool calls, evidence, risk score, decision rule, approval, and customer notice. The audit trail is 15 points, and it is easy to forget when time is tight.
- Ground every remedy in a versioned policy file and check eligibility against the facts of the case. An arbitrary refund loses points even when the customer would be happy with it.
- Be honest in the pitch about what is real and what is mocked. Overclaiming costs you on Demo clarity, and a crisp, honest narrative scores higher than a flashy one.

LOGISTICS

- Final: Friday 12 June 2026, Abu Dhabi. Reception opens at 8:00 AM.
- Bring your laptop with the serial number you registered.
- Teams present to the judging panel: a 5-minute live technical demo of both journeys, plus the visible audit trail.
- Use mock data only. No live financial systems.